

Tan Ngo

+84 364-143-275 — Hanoi, Vietnam

[✉ tanngo06022002@gmail.com](mailto:tanngo06022002@gmail.com) — [🌐 Personal Page](#) — [🐙 GitHub](#)

EDUCATION

Hanoi University of Science and Technology (HUST)

Aug 2021 – Apr 2025

B.Sc. in Computer Science, Hanoi, Vietnam (GPA: 3.69/4.0)

- Graduated early in **3.5 years**; one of 9 outstanding graduating students out of ~3,000 (Director's Certificate of Merit).
- Selected to deliver the **graduation speech** on behalf of the class.

Lam Son High School for the Gifted — *Specialized Mathematics Program, Vietnam*

Aug 2018 – May 2021

- **Vietnam National Mathematics Competition, Second Prize** (2021): 11th nationally among 400 top students.

RESEARCH INTEREST

AI Research Resident on the Computer Vision team at [Qualcomm AI Research Vietnam](#) (advised by [Phong Ha Nguyen](#), [Binh-Son Hua](#), and [Hung H. Bui](#)), building controllable **video diffusion models** and **world models** for **robot-learning** data generation. Broadly interested in **efficient ML**, **parallel and distributed computing**, and **ML systems**.

PUBLICATIONS

UniCaMo: Track the Noise, Move the World: 3D-Grounded Motion-Consistent Noise for Controllable Video Generation

- Long Vu*, **Tan Ngo***, Animesh Karnewar, Amir Habibian, Binh-Son Hua, Hung H. Bui, Minh Hoai, and Phong Ha Nguyen. *Under review at NeurIPS 2026*. · ***Co-first author** (equal contribution). · [Link](#)

RESEARCH EXPERIENCE

AI Research Resident, Qualcomm AI Research Vietnam

Jun 2025 – Present

– **UniCaMo - Unifying Camera and Motion Control for Video Generation** (Jan 2026 – Present)

- Building a video diffusion model that jointly controls **camera** and **object motion** in **3D space**, using explicit depth as a 3D signal for cross-frame motion consistency.
- Scaling training/inference across **multi-node GPU clusters (up to 300 GPUs)** on RunAI.
- Fixed a **DiffSynth-Studio** loading bug that cut per-node startup 9 min → ~44 s and removed a recurring multi-node crash.

– **Self-Directed Systems Research - Inspired from UniCaMo paper** ([Link](#)) (2026 – Present)

- Sped up my own **Ray**-based evaluation pipeline **6.7×** (397.3 s → 59.7 s on an 8× H100 node) by fixing a CPU-thread default that left every worker single-threaded despite full hardware bandwidth.
- Found that adding GPUs can *slow* a pipeline: evaluation ran fastest on **2 of 8 GPUs**, depth/pose annotation on **4 of 8**, once the container's fixed CPU budget became the bottleneck.

– **Few-Step Text-to-Multiview Diffusion Model** (Jun 2025 – Dec 2025)

- **Distilled** a multi-step MVDream text-to-multiview model into a **few-step** student at **512×512** (4× the teacher's resolution).
- **Quantized** the model and shipped it on-device on Samsung S24; demoed at **Qualcomm's NeurIPS 2025** booth.

Research Assistant, Computer Vision Lab, BKAI, HUST

Oct 2023 – Jan 2025

– **Continual Learning for Object Detection with Generative Data Replay** ([Thesis-Link](#)) (Sep 2024 – Jan 2025)

- Applied continual learning to object detection, using diffusion-generated replay data to mitigate catastrophic forgetting without storing past-task data.

– **Vision Transformers for Medical Endoscopy Image Segmentation** (Oct 2023 – Aug 2024)

- Researched ViT-based segmentation of medical endoscopy images using MAE self-supervised pretraining on unlabeled data.

WORK EXPERIENCE

- Generative AI Engineer, ZenAI Aug 2024 – Jun 2025
- **Diffusion Image-Generation Acceleration** (Aug 2024 – Jun 2025)
 - **Sped up** production diffusion inference via parallelism + `torch.compile`: FLUX.1-schnell 2.02 s $\rightarrow$ 0.86 s ($\approx 2.35\times$), SDXL Turbo 1.26 s $\rightarrow$ 0.72 s ($\approx 1.75\times$), while keeping image quality.
 - **Fixed a `torch.compile` bug** (v2.3.1) that was silently blocking these speedups.
 - Diagnosed a **dynamic-shape problem**: `torch.compile`'s speedup holds for only ~ 16 distinct input shapes before recompilation erases it; fixed via shape bucketing.
 - **LLM Serving & Inference Optimization (vLLM, SGLang)** (Aug 2024 – Jun 2025)
 - Served and optimized LLMs with **vLLM/SGLang**: applied **quantization**, **KV-cache** tuning, and **parallel GPU** execution to cut **latency** and raise **throughput** in production.

HONORS AND AWARDS

- WISEMI Take-off Fellowship** — Selective talent program supporting Vietnamese STEM students pursuing PhD programs in top countries. 2026
- HUST Wings Scholarship** — Awarded for 6 consecutive semesters. 2022–2024
- HUST Academic Encouragement Scholarship** — Computer Science department, $<3\%$ selected. 2021–2022
- Vietnam National Mathematics Competition, Consolation Prize** — National award in Vietnam's highest-level mathematics competition for gifted high school students. 2020
- North Central Region Mathematics Olympiad, First Prize** — Among specialized high schools in Vietnam. 2019

TECHNICAL SKILLS

- Languages:** Python, C/C++
- Distributed & parallel:** multi-GPU / multi-node training and inference (up to 300 GPUs), NCCL, CPU/GPU workload balancing, Ray, SLURM
- Efficiency & compilers:** `torch.compile`, quantization, KV-cache, distillation, vLLM, SGLang
- Frameworks & libraries:** PyTorch, Diffusers, ComfyUI, NumPy, OpenCV

REFERENCES

- Dr. Binh-Son Hua** — Assistant Professor, Trinity College Dublin; Consultant, Qualcomm AI Research Vietnam
E-mail: binhson.hua@tcd.ie
- Dr. Phong Ha Nguyen** — Senior Research Scientist, Qualcomm AI Research Vietnam
E-mail: phongnh@qti.qualcomm.com
- Dr. Hung H. Bui** — VP, Qualcomm; ex-Founder & CEO, VinAI; ex-DeepMind
E-mail: hungbui@qti.qualcomm.com